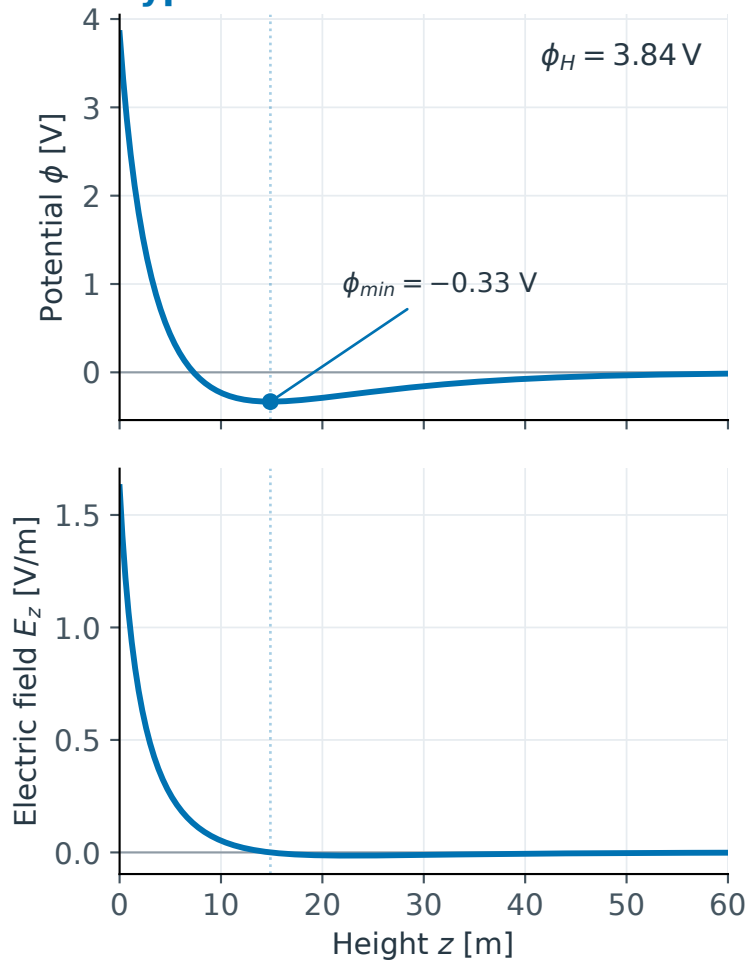


Representative 1D sheath solutions

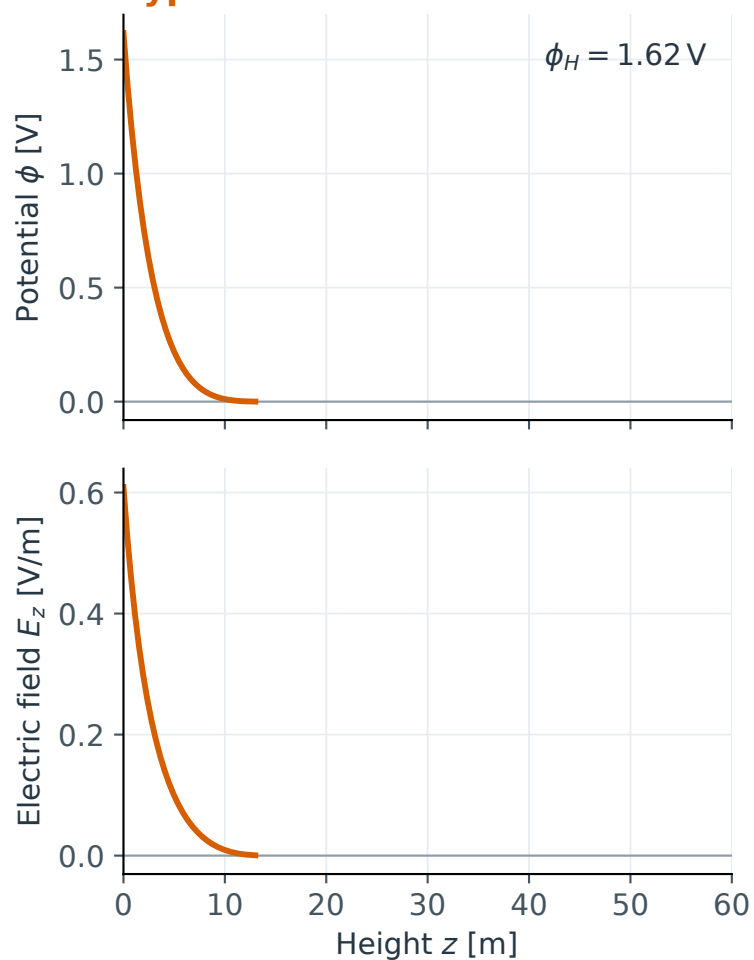
$n_i = 8.7 \text{ cm}^{-3}$ | $T_e = 12 \text{ eV}$ | $T_{pe} = 2.2 \text{ eV}$ | $v_{sw} = 468 \text{ km s}^{-1}$ | $u_e = 0$

Zero current $J_z = 0$ | $n_{pe,ref} = 64 \text{ cm}^{-3}$ | $v_i = v_{sw} \sin \alpha$, $n_{pe,0} = n_{pe,ref} \sin \alpha$

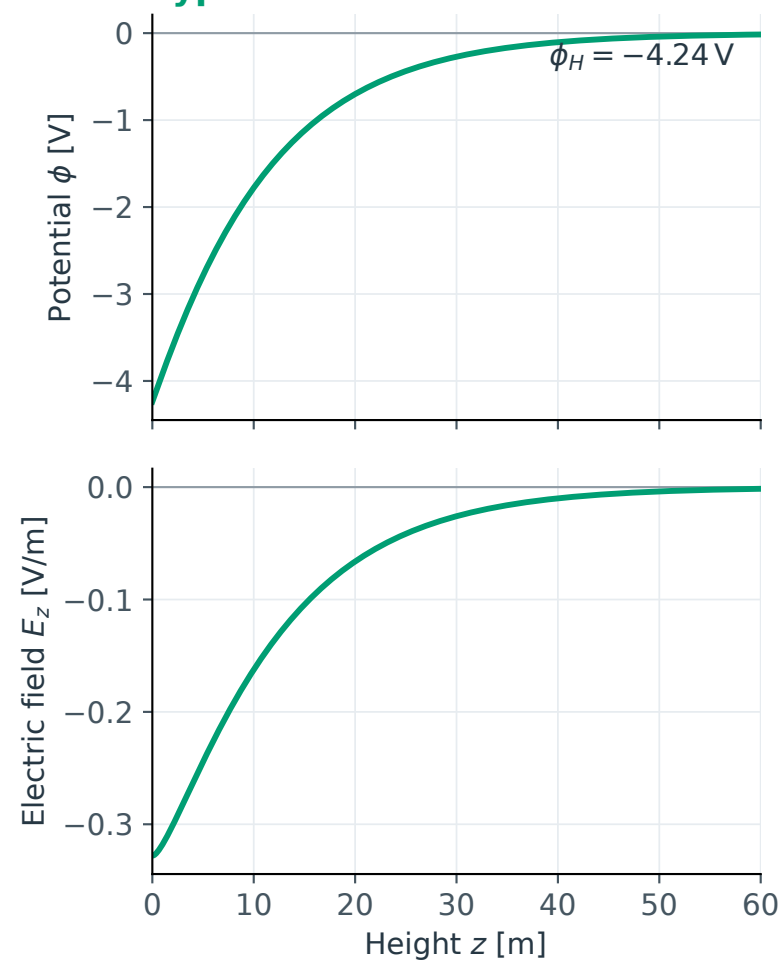
Type A · $\alpha = 60^\circ$



Type B · $\alpha = 20^\circ$



Type C · $\alpha = 10^\circ$



+z: surface $\rightarrow$ upstream. $E_z = -d\phi/dz$. All profiles approach zero potential and field upstream; the first 60 m are shown.